

## Computer Vision & AI Intern (OpenCV / YOLO) – Final Year Engineering Students

**Location:** Chennai - On-site Only

**Internship Duration:** 6 to 12 Months

**Company:** Ventura Automation Services / FIDAS (Fabric Inspection & Defect Analysis System)

### Computer Vision & AI Internship

Join our engineering team in developing an advanced **AI-powered Fabric Inspection System** for the textile industry. The project combines **Industrial Cameras, Embedded AI, Computer Vision, Deep Learning, and Edge Computing** to detect weaving and knitting defects in real time.

The system is designed using industrial cameras, Rockchip AI processors, embedded Linux, OpenCV, and YOLO models for deployment in textile mills.

This internship offers an opportunity to work on a real industrial product rather than an academic project.

### Responsibilities

The intern will be involved in

- Developing image processing algorithms using OpenCV
- Training and optimizing YOLO models for textile defect detection
- Dataset collection, annotation and augmentation
- Camera calibration, image acquisition, Image preprocessing and enhancement
- Industrial lighting evaluation and optimization
- Designing inference pipelines for embedded AI devices
- Performance optimization for real-time detection
- Model evaluation using Precision, Recall and mAP
- Working with industrial monochrome and global shutter cameras
- Assisting in GUI integration and inspection software
- Preparing technical documentation
- Testing systems in real production environments

### Required Technical Skills

#### Computer Vision

- OpenCV (Mandatory)
- Image Processing
- Feature Extraction
- Contour Detection & Edge Detection
- Perspective Transformation & Camera Calibration

## **Deep Learning**

- YOLO (Mandatory)
- Object Detection, Instance Segmentation & Dataset Preparation
- Annotation Tools, Model Training & Model Evaluation
- Hyperparameter Tuning

## **Programming**

- Python (Mandatory)
- NumPy
- Pandas
- Git

## **AI Frameworks**

Experience in one or more

- PyTorch, TensorFlow
- ONNX or Ultralytics YOLO

## **Nice to Have**

- Industrial Cameras
- Embedded Linux, Raspberry Pi or Rockchip RV1106
- GStreamer, Linux Shell

## **Eligibility**

- Final Year B.E./B.Tech/M.E./M.Tech students
- Artificial Intelligence
- Electronics & Communication
- Electrical & Electronics
- Robotics
- Embedded Systems

Students from other relevant disciplines with strong computer vision skills are also encouraged to apply.

## **Preferred Candidate Profile**

We are looking for students who

- Are passionate about Computer Vision and AI
- Have built personal or academic OpenCV/YOLO projects
- Enjoy solving challenging engineering problems
- Can work independently with minimal supervision
- Have strong analytical and debugging skills
- Are willing to work on industrial automation products

## Internship Duration

- Minimum: **6 Months**
- Preferred: **9-12 Months**
- Full-time internship preferred.

## What You'll Learn

- Industrial Machine Vision & AI-based Defect Detection / Classification
- Embedded AI Development
- Edge Computing
- Industrial Camera Integration
- Deep Learning Model Optimization & Real-Time Inference
- Industrial Automation
- Production-grade Software Development
- Product Development Lifecycle

## Deliverables

Interns will be expected to

- Develop production-quality computer vision modules
- Train high-accuracy YOLO models
- Improve detection accuracy and inference speed
- Maintain proper documentation
- Present weekly progress reports
- Demonstrate completed modules at project milestones

## Performance Evaluation

Interns will be evaluated based on

- Technical competency
- Code quality
- Innovation
- Learning ability
- Team collaboration
- Documentation
- Attendance and professionalism
- Timely completion of assigned milestones

Outstanding interns may be considered for a **full-time position** upon successful completion of the internship.

## Selection Process

1. Technical Assessment (OpenCV & Python)
2. Practical Coding Challenge & Technical Interview
3. Final Discussion

## Stipend

- **Competitive stipend** based on technical skills and interview performance.
- Additional incentives may be offered for exceptional contributions to product development.

## Bonus Skills (Highly Preferred)

Candidates with experience in any of the following will have an advantage:

- YOLOv8/YOLOv11 customization
- OpenCV optimization
- Industrial camera SDKs
- MVTec AD or textile defect datasets
- OCR and barcode recognition
- Embedded AI deployment (RKNN, TensorRT, ONNX Runtime)
- Linux application development
- Industrial Ethernet (GigE Vision) and MIPI CSI camera interfaces

This role is ideal for students seeking substantial industry experience in AI and computer vision while contributing to the development of a commercial-grade industrial inspection system.